

Control work 9

— data handling

Frequency tables, the three averages, and pie charts drawn and read.

LESSON 167

1 × 40 MIN

MATEMATIKA 6, NAZORAT ISHI 9

S7 6 AND 16 REVIEW

LESSON CLOCK

2'

28'

8'

2'

Instructions	2 min
The paper	28 min
Answers and diagnosis	8 min
What comes next	2 min

LEARNING OBJECTIVES

- Complete a frequency table and find its total.
- Find the mean, median, mode and range from a table.
- Turn a frequency into a sector angle and back again.
- Name each lost mark by its type and rewrite the whole solution.

TEXTBOOK

National	pp. 449–479
Cambridge	Stage 7, pp. 58–184
Workbook	Control paper 9

01

Explanation

The paper — 20 marks, 28 minutes

Questions 1 to 4 use this table of marks out of ten for one class.

MARK	5	6	7	8
frequency	3	6	7	4

Q	TASK	MARKS	FROM
1	The number of pupils in the class	2	L157
2	The mode and the range	3	L157
3	The median	3	L157
4	The mean	3	L157
5	A survey of 45 pupils: the sector angle for a group of 10	3	L160–163
6	A pie chart of 200 people has a 63° sector: how many people?	3	L164–166
7	Express 108° as a percentage of the whole	2	L164–166
8	Give one reason a survey of the school football team is biased	1	L158–159

THE ANSWERS

20 pupils; mode 7, range 3; median 7; mean 6.6; 80° ; 35 people; 30%; the team already prefers football.

Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
the mean divided by the number of marks	$132 \div 4$	$132 \div 20$
the range given as an interval	5 to 8	3
the median taken as the middle mark of the row	6.5	7
the angle taken as the frequency over the total	$10 \div 45$	$10 \cdot 8 = 80^\circ$
the sector read as the number of people	63	35
the percentage division reversed	$3.6 \div 108$	$108 \div 3.6$
the degree sign missing	80	80°

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

What comes next

IF YOU LOST MARKS ON	REVISE
Q1–Q4	frequency tables and the averages, L157
Q5	drawing a pie chart, L160–163
Q6–Q7	reading a pie chart, L164–166
Q8	samples and bias, L158–159

LOOKING FORWARD

Next comes the choice of chart for a given kind of data, then graphs of relationships, probability, the transformations on a grid, and the three-dimensional shapes.

02 Worked examples

EXAMPLE 1 Model answer, Q3: the median of the twenty marks.

- 1 With 20 values the median is between the 10th and 11th.
- 2 Cumulative frequencies: 3, 9, 16, 20.
Both fall in the mark 7.
- 3 The median is 7.

ANSWER

7

EXAMPLE 2 Model answer, Q5: 45 pupils, a group of 10.

① $360 \div 45 = 8^\circ$ for each pupil.
What one person is worth.

② $10 \cdot 8.$

③ $= 80^\circ.$
Under a quarter of the circle ✓

ANSWER 80° **EXAMPLE 3** Model answer, Q6: a 63° sector of 200 people.

① $\frac{63}{360}$ of the survey.

② $\frac{63}{360} \cdot 200.$

③ $= 35$ people.
Under a fifth of 200 ✓

ANSWER

35 people

03 Interactive model

Return Q4 first and ask who divided by four; naming it as “dividing by the number of different marks” fixes it permanently.

The paper in eight questions

Frequencies 3, 6, 7, 4 give a class of:

The mode of that table is:

The range is:

The median is:

6

6.5

7

7.5

The mean is:

6.6

33

7

6.5

With 45 pupils, 10 of them take:

45°

80°

10°

36°

A 63° sector of 200 people is:

63

35

50

20

108° as a percentage is:

10.8%

30%

36%

108%

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Frequency table	Chastotalar jadvali	Таблица частот
Mean	O‘rtacha arifmetik	Среднее арифметическое
Median	Mediana	Медиана
Mode	Moda	Мода
Range	Farq (kenglik)	Размах
Sector angle	Sektor burchagi	Угол сектора
Bias	Yon bosish	Смещение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Q1 is answered by:

adding the marks

adding the frequencies

counting the columns

dividing

Q4 divides the total of the marks by:

4

20

132

8

Q5 first finds:

the percentage

what one pupil is worth

the total angle

the median

Q6 needs:

the angle only

the angle and the total

the total only

the mode

Q7 needs no total because a percentage is:

a number

a share

an angle

a frequency

Q8 is about:

arithmetic

a biased sample

the mean

a chart

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The total of frequencies 3, 6, 7, 4

ANSWER 20

2. The mode of that table

ANSWER 7

3. The range of marks from 5 to 8

ANSWER 3

4. 45 pupils: the angle for one

ANSWER 8°

5. 10 pupils out of 45

ANSWER 80°

6. 108° as a percentage

ANSWER 30%

7. The angles of a pie chart total

ANSWER 360°

Medium · the standard question

7 PROBLEMS

1. The median of that table

ANSWER 7

2. The mean of that table

ANSWER 6.6

3. The total of the twenty marks

ANSWER 132

4. 200 people, a sector of 63°

ANSWER 35 people

5. How many scored 7 or more

ANSWER 11

6. How many scored below 7

ANSWER 9

7. The frequency of the mark 6

ANSWER 6

Hard · stretch and reason

7 PROBLEMS

1. Why is $132 \div 4$ the wrong mean?

ANSWER It divides by the number of different marks

2. 45 pupils, a group of 18: the angle

ANSWER 144°

3. 200 people, a sector of 90°

ANSWER 50 people

4. A 21st pupil scores 8: the new mean, to 2 d.p.

ANSWER 6.67

5. A survey of the football team gives

ANSWER Too large a share for football

6. 90° as a percentage

ANSWER 25%

7. The median of an even number of values is

ANSWER Halfway between the two middle values

07 Homework

Homework — 5 tasks

Rewrite every question you lost a mark on in full, with the units and the degree signs.

1. Rewrite in full every question on which you lost a mark, naming the slip in the margin.
2. Marks 4, 5, 6, 7 have frequencies 2, 5, 8, 5. Find the mean, median, mode and range.
3. A survey of 90 pupils has a group of 25. Find the sector angle.
4. A pie chart of 300 people has a 126° sector. How many people is that?
5. Express 72° as a percentage and as a fraction of the whole.